

MIP_WAVE_2_EXECUTION_PLAN.md

Mainframe Intelligence Platform

Wave 2 – Engineering Foundation

Version: 1.0

Author: MIP Core Team

Status: In Progress

Executive Summary

Wave 2 establishes the engineering foundation required to build the Mainframe Intelligence Platform (MIP).

The purpose of Wave 2 is to create a repeatable, scalable, and AI-assisted engineering environment that enables teams to build MIP consistently across organizations.

Wave 2 does not focus on parsing COBOL, building graphs, or implementing AI capabilities.

Wave 2 focuses on establishing the engineering operating model.

At the completion of Wave 2, every engineer should be able to:

- Clone the repository
 - Configure the workspace
 - Use GitHub Copilot consistently
 - Follow engineering standards
 - Use reusable skills
 - Use reusable prompts
 - Follow testing standards
 - Contribute without tribal knowledge
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Wave 2 Objectives

Primary Objectives

Establish:

- Workspace Standards
- Repository Standards

- Copilot Standards
 - Skill Framework
 - Prompt Framework
 - Coding Standards
 - Testing Standards
 - Engineering Playbook
-

Deliverables

Wave 2 consists of eight deliverables.

Deliverable 1

MIP_WORKSPACE_BOOTSTRAP.md

Purpose:

Provide a complete onboarding guide.

Outcome:

Any engineer can become productive within one day.

Deliverable 2

MIP_REPOSITORY_STRUCTURE.md

Purpose:

Define repository organization and ownership.

Outcome:

Consistent repository layout across teams.

Deliverable 3

MIP_GITHUB_COPILOT_STRATEGY.md

Purpose:

Define how GitHub Copilot is used throughout MIP.

Outcome:

Consistent AI-assisted development workflow.

Deliverable 4

Skills Framework

Purpose:

Create reusable Copilot skills.

Outcome:

Consistent engineering behavior.

Required Skills:

- repository-engineer
 - mainframe-modernization-architect
 - mainframe-code-analyst
 - metadata-modeler
 - graph-engineer
 - sqlite-engineer
 - test-engineer
 - code-reviewer
 - documentation-writer
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Deliverable 5

Prompt Library

Purpose:

Create reusable engineering prompts.

Outcome:

Consistent prompt-driven workflows.

Libraries:

- architecture
- discovery
- parsing
- metadata
- graph
- testing
- review

- modernization
-

Deliverable 6

MIP_CODING_STANDARDS.md

Purpose:

Define engineering standards.

Outcome:

Consistent implementation quality.

Deliverable 7

MIP_TESTING_STRATEGY.md

Purpose:

Define testing standards.

Outcome:

Reliable and maintainable codebase.

Deliverable 8

MIP_ENGINEERING_PLAYBOOK.md

Purpose:

Document the engineering operating model.

Outcome:

Repeatable delivery process.

Engineering Principles

All Wave 2 artifacts must follow these principles.

Principle 1

Metadata First

Everything begins with metadata.

Never design around AI.

Always design around metadata.

Principle 2

Relationships Drive Understanding

Programs alone do not create understanding.

Relationships create understanding.

Examples:

Program → Program

Program → Copybook

Program → Table

Job → Program

Principle 3

Knowledge Before Modernization

Never modernize what is not understood.

Sequence:

Discovery

↓

Metadata

↓

Relationships

↓

Knowledge Graph

↓

Reasoning

↓

Modernization

Principle 4

AI Is An Accelerator

AI is not the source of truth.

Source code remains the source of truth.

Copilot assists engineering.

It does not replace engineering.

Delivery Sequence

Wave 2 must be executed in the following order.

Phase A

Workspace Foundation

Deliverables:

- MIP_WORKSPACE_BOOTSTRAP.md
- MIP_REPOSITORY_STRUCTURE.md

Outcome:

Standardized workspace.

Phase B

AI-Assisted Engineering

Deliverables:

- MIP_GITHUB_COPILOT_STRATEGY.md

Outcome:

Standardized Copilot usage.

Phase C

Skill Framework

Deliverables:

All skill.md files.

Outcome:

Reusable engineering expertise.

Phase D

Prompt Framework

Deliverables:

Prompt library.

Outcome:

Reusable engineering workflows.

Phase E

Engineering Governance

Deliverables:

- MIP_CODING_STANDARDS.md
- MIP_TESTING_STRATEGY.md
- MIP_ENGINEERING_PLAYBOOK.md

Outcome:

Consistent engineering practices.

Repository Layout

The following structure must be used.

mip-platform/

.github/

skills/

prompts/

docs/

src/

tests/

output/

scripts/

tools/

examples/

data/

logs/

Skills Directory Structure

skills/

repository-engineer/

skill.md

mainframe-modernization-architect/

skill.md

mainframe-code-analyst/

skill.md

metadata-modeler/

skill.md

graph-engineer/

skill.md

sqlite-engineer/

skill.md

test-engineer/

skill.md

code-reviewer/

skill.md

documentation-writer/

skill.md

Prompt Library Structure

prompts/

architecture/

discovery/

parsing/

metadata/

graph/

testing/

review/

modernization/

Success Metrics

Wave 2 is successful when:

A new engineer can:

- Clone repository
- Install dependencies
- Configure workspace
- Use skills
- Use prompts
- Run tests
- Follow standards

Without requiring assistance from project maintainers.

Risks

Risk 1

Overengineering

Mitigation:

Keep solutions simple.

Risk 2

Prompt Sprawl

Mitigation:

Centralized prompt library.

Risk 3

Skill Duplication

Mitigation:

Clear ownership and responsibilities.

Risk 4

Copilot Inconsistency

Mitigation:

Workspace instructions and reusable skills.

Definition Of Done

Wave 2 is complete when:

- ✓ Workspace Bootstrap completed
- ✓ Repository Structure completed
- ✓ GitHub Copilot Strategy completed
- ✓ Skills Framework completed
- ✓ Prompt Library completed
- ✓ Coding Standards completed
- ✓ Testing Strategy completed
- ✓ Engineering Playbook completed
- ✓ Documentation reviewed
- ✓ Repository baseline tagged

Tag:

v0.2-engineering-foundation

Exit Criteria

Upon completion of Wave 2, the team may proceed to Wave 3.

Wave 3 introduces:

- Metadata Model
- SQLite Schema

- NetworkX Model
- Knowledge Graph Design
- TSys Discovery Playbook
- Phase 0–4 Implementation Specifications

These become the foundation for building the first operational version of MIP.